CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The kidneys in human beings are a part of the system for: 1

(a) nutrition **(b)** excretion
(c) respiration **(d)** transportation of oxygen

2. Aerobic respiration in humans yields: 1

(a) ethanol and CO_2 **(b)** only energy
(c) lactic acid **(d)** CO_2, water and energy

3. The functional unit of the kidney is: 1

(a) villus **(b)** nephron **(c)** neuron **(d)** alveolus

4. The xylem in plants is responsible for: 1

(a) transport of oxygen **(b)** transport of food
(c) transport of water **(d)** transport of amino acids

5. In muscles during vigorous exercise, glucose breaks down anaerobically to give: 1

(a) CO_2 only **(b)** water **(c)** ethanol **(d)** lactic acid

6. Urine is stored temporarily in: 1

(a) kidney **(b)** ureter **(c)** urethra **(d)** urinary bladder

7. Which of the following are required for autotrophic nutrition in green plants? 1

(a) only chlorophyll and sunlight
(b) carbon dioxide, water, chlorophyll and sunlight
(c) only carbon dioxide and water
(d) only sunlight

8. In yeast, anaerobic respiration produces: 1

(a) water **(b)** glucose
(c) lactic acid **(d)** ethanol and carbon dioxide

9. The filtration of blood in the nephron takes place in: 1

(a) tubule **(b)** glomerulus **(c)** collecting duct **(d)** ureter

10. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in: 1

(a) nucleus **(b)** cytoplasm **(c)** mitochondria **(d)** chloroplast

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What are the differences between autotrophic nutrition and heterotrophic nutrition? 2

12. How is the small intestine designed to absorb digested food? 2

13. What are the components of the transport system in highly organised plants? 2

14. Where do plants get each of the raw materials required for photosynthesis? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is the process of respiration considered an exothermic process? Write its equation. 3

16. Describe the working of the nephron in forming urine. 3

17. Compare aerobic and anaerobic respiration with one product each. 3

OR

How do guard cells regulate the opening and closing of stomatal pores? 3

18. Why are the alveoli covered with a network of blood vessels? 3

OR

Why do we feel cramps in muscles after vigorous exercise? Name the substance. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the steps of photosynthesis: absorption of light, splitting of water, reduction of CO_2. What happens in desert plants at night? 5

OR

Describe excretion in human beings: parts of the system and the working of the nephron. 5

20. Describe the human alimentary canal with the role of each part. 5

OR

Describe different types of nutrition in organisms with examples: autotrophic, parasitic, saprophytic and holozoic. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND